

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

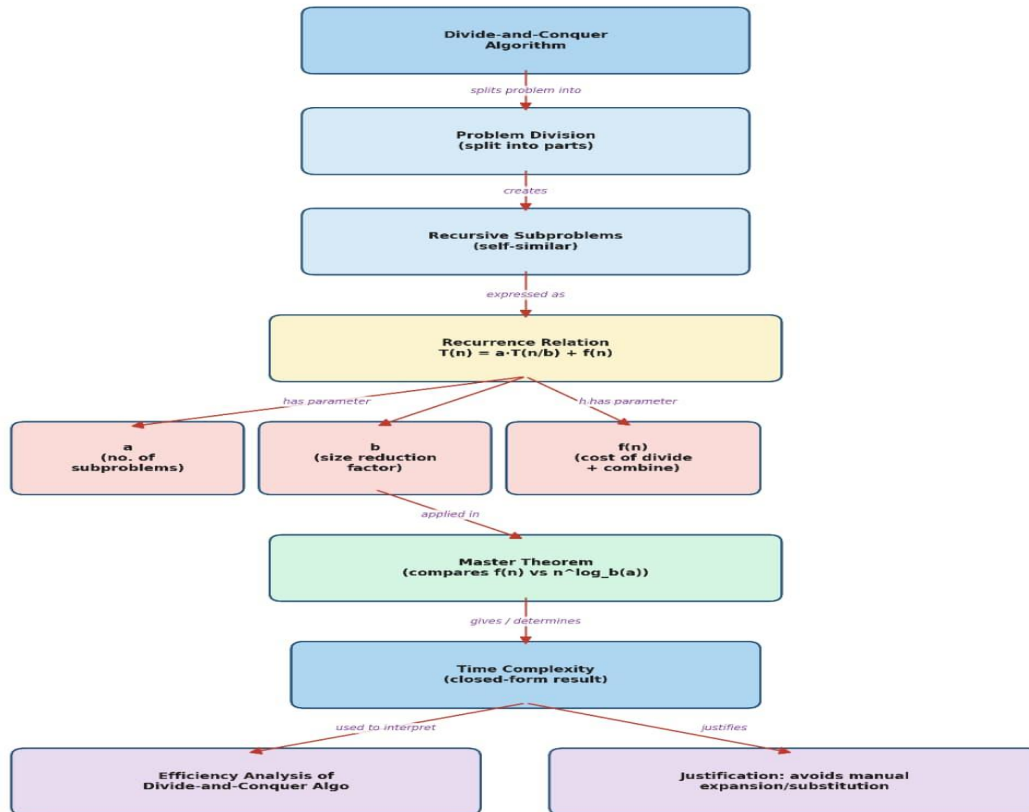
- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – CONCEPT MAPPING

Q1. Explain how dividing a problem leads to recursive subproblems and recurrence formulation. Extend the map by including parameters a, b, and f(n) under recurrence relation. Justify how Master Theorem simplifies solving such recurrences. Interpret how this map helps analyze divide-and-conquer algorithms efficiently.

Concept Map: Divide-and-Conquer → Recurrence Relation → Master Theorem



- Divide-and-conquer algorithm: a technique where a large problem is split into smaller, more manageable sub-problems that are solved individually
- Problem division: the input is logically split into equal or near-equal parts (e.g., splitting an array into 2 halves)
- Recursive subproblems: each sub-problem has the same structure as the original problem, so the same function calls itself recursively
- Recurrence relation $t(n) = a \cdot t(n/b) + f(n)$: the mathematical way of representing this recursive process
- A: the number of sub-problems generated (branching factor)
- B: the factor by which each sub-problem's size shrinks
- F(n): the extra cost of dividing the problem and combining the results
- Master theorem: compares f(n) with $n^{\log_b a}$ and gives the time complexity directly using 3 standard cases — no need to manually solve the recurrence
- Time complexity: the final result showing how efficient the algorithm is
- Efficiency analysis / justification: this entire flow lets us quickly analyze any divide-and-conquer algorithm (merge sort, quick sort, binary search) without manual expansion

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____